

Force Myography Compression Study Summary

Eshaan Modi | 9th grade | eshaanmodi541@gmail.com

Project Statement

Force myography's main limitation is cross-session variance. In the papers I've found, cross-session sensor drift is handled with recalibration or domain adaptation rather than by controlling the mechanical variable directly. This project aims to fix this by correcting band compression – a variable that can affect gesture classification accuracy between don/doff cycles – in hardware.

When a band is re-donned it struggles from two mechanical confounds; namely band placement and compression. Finding the effect of band compression on sensor drift not only removes a potential source of classification accuracy loss, it also removes one of the two confounding signals thus allowing us to better quantify the components of sensor drift in FMG.

Approach

The modified band uses a pneumatic compression system: an inflatable sac constrained by a rigid outer ring, so inflation drives pressure inward onto the sensors against the skin. An embedded pressure sensor closes the feedback loop, letting the band hold a commanded pressure and return to a per-participant baseline recorded in the first session. Two sensor types are instrumented in parallel — force-sensitive resistors, the field standard, and barometric touch sensors, a newer approach — to test whether one is inherently less sensitive to compression variation. Placement is held constant independently of compression using anatomical landmarking and a keyed alignment feature on the band, so that drift in Stage 2 can be attributed to compression rather than repositioning.

The experiment runs in two stages. First, a within-session pressure sweep measuring classification accuracy across pressure setpoints for both sensor types. Cho et al. found a rise-then-plateau across three discrete belt tensions; under continuous control I want to locate where that plateau begins and whether it differs between the two sensor types. A flat response would indicate placement dominates drift – still a useful negative result. Second, a cross-session comparison over multiple days with ten participants, running a conventional fixed strap against the dynamic band regulating each participant's session-one pressure. Training on the first session and testing on later ones isolates the effect of compression control on cross-session drift.

Current Status & Timeline

As of now, I am conducting deeper literature reviews and starting designing the project CAD and PCB files. I hope to have finished project design by mid-August, a revised prototype by October, preliminary data by December, and a completed project by February as this project will ultimately be presented at the Chester County Science Research Fair – a feeder into ISEF – in March. I hope a mentor can help me review my design, give me statistical guidance, sanity-check my protocol, and help me condense/analyze my findings.

Core Bibliography

Xiao, Zhen Gang, and Carlo Menon. "A review of force myography research and development." *Sensors* 19.20 (2019): 4557.

Sherif, Omar, Mohamed Mahgoub Bassuoni, and Omar Mehrez. "A survey on the state of the art of force myography technique (FMG): analysis and assessment." *Medical & Biological Engineering & Computing* 62.5 (2024): 1313-1332.

Jiang, Xianta, et al. "Exploration of force myography and surface electromyography in hand gesture classification." *Medical engineering & physics* 41.1 (2017): 63-73.

Cho, Sung-Gwi, Hirochika Matsui, and Kengo Ohnishi. "Accuracy Comparison of FMG-based Hand-wrist Gesture Recognition with and without Compression." *IEEE Access* (2026).

Marquardt, Charlotte, et al. "Embedded barometric pressure sensor unit for force myography in exoskeletons." *2022 IEEE-RAS 21st International Conference on Humanoid Robots (Humanoids)*. IEEE, 2022.